## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 1 of 7 · Question 7 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 7 | (a) (i) | • when both present;<br>• both (equally) expressed; | | 2 |
| | (ii) | • $I^BI^O$;<br>• $I^BI^B$; | | 2 |
| | (b) | any four from<br>• carried on X chromosome;<br>• not expressed in heterozygotes/recessive;<br>• can't have male carriers;<br>• can only have female heterozygotes/carriers;<br>• because only females can have 2X chromosomes<br>• male XY/only one X chromosome; | | 4 |

Total 8 marks

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 1 of 7 · Question 7 part b · 4 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 2 of 7 · Question 6 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 6 | (a) (i) | any three from<br>• prevents excessive loss of blood (1)<br>• forms scab over wound (1)<br>• prevents entry of bacteria /pathogens/microorganisms(1)<br>• so prevents infection (1)<br>• allows wound to heal (1) | | 3 |
| | (ii) | • (allele) on X/sex chromosome (1)<br>• passed on to offspring/inherited by children (1) | | 2 |
| | (b) (i) | D; ($X^hY$)<br>A the genotype is not $X^hX^h$<br>B the genotype is not $X^HX^h$<br>C the genotype is not $X^HY$ | | 1 |
| | (ii) | B; ($X^HX^h$)<br>A the genotype is not $X^h X^h$<br>C the genotype is not $X^HY$<br>D the genotype is not $X^hY$ | | 1 |
| | (iii) | • offspring genotypes $X^HX^h$ and $X^hY$ (1)<br>• 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1) | full marks for correct figures only. | 2 |
| | (iv) | • unusual for female to be haemophiliac (1)<br>• both alleles have to be affected (1)<br>• must have a haemophiliac father and mother with haemophilia/carrier (1) | any correct reference to excessive blood loss | 3 |

Total for question = 12 marks

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 2 of 7 · Question 6 part a.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 3 of 7 · Question 6 part b.iv · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.iv in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 6 (a) | (i) | any three from<br>• prevents excessive loss of blood (1)<br>• forms scab over wound (1)<br>• prevents entry of bacteria /pathogens/microorganisms(1)<br>• so prevents infection (1)<br>• allows wound to heal (1) | | 3 |
| | (ii) | • (allele) on X/sex chromosome (1)<br>• passed on to offspring/inherited by children (1) | | 2 |
| (b) | (i) | D; ($X^hY$)<br>A the genotype is not $X^hX^h$<br>B the genotype is not $X^HX^h$<br>C the genotype is not $X^HY$ | | 1 |
| | (ii) | B; ($X^HX^h$)<br>A the genotype is not $X^h X^h$<br>C the genotype is not $X^HY$<br>D the genotype is not $X^hY$ | | 1 |
| | (iii) | • offspring genotypes $X^HX^h$ and $X^hY$ (1)<br>• 1 carrier female to 1 haemophiliac male = 50%/0.5/½ (1) | full marks for correct figures only. | 2 |
| | (iv) | • unusual for female to be haemophiliac (1)<br>• both alleles have to be affected (1)<br>• must have a haemophiliac father and mother with haemophilia/carrier (1) | any correct reference to excessive blood loss | 3 |

Total for question = 12 marks

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 3 of 7 · Question 6 part b.iv · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 4 of 7 · Question 8 part c · 5 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 8 (a) | any three from<br>• change in base sequence/named mutation (1)<br>• of DNA (1)<br>• causing change in genotype/gene (1)<br>• incorrect protein produced (1)<br>• change in phenotype/characteristics (1) | Allow change in DNA for mps 1 and 2 (1) | 3 |
| (b) (i) | father $X^HY$ mother $X^HX^h$ (1)<br>gametes $X^H$ Y $X^H$ $X^h$ (1)<br>F1 $X^HX^H$ normal female; $X^HX^h$ carrier female; $X^HY$ normal male; $X^hY$ haemophiliac(1) male | | 4 |
| (ii) | (chance of first and second child is) ¼/0.25/25% (1)<br>for both is ¼ x ¼/0.25 x 0.25 (1)<br>= 1/16/0.0625/6.25% (1) | | 3 |
| (c) | any five from<br>• males only have one X chromosome (1)<br>• two X chromosomes/H and h needed to be a carrier (1)<br>• recessive allele/h (1)<br>• carried on X chromosome (1)<br>• either present or not (1)<br>• no chance for dominant allele to be present (1)<br>• to mask recessive (1) | | 5 |

Total 15 marks

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 4 of 7 · Question 8 part c · 5 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain sex-linked inheritance and why males are affected more

Answer 5 of 7 · Question 9 part b.i · 2 marks

2024 JUN · Paper 01 · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 9 (a) | (i) | A, B, AB; | 2 marks for all 3 correct, 1 mark for 1 or 2 correct | (2) |
| | (ii) | any five from<br>• blood type O;<br>• no antigens present;<br>• (person) contains anti-A and anti-B antibodies;<br>• other blood groups have A/B antigens;<br>• immune response if mixed;<br>• causes agglutination of red blood cells; | | (5) |
| (b) | (i) | • (faulty allele) carried/found on female sex/X chromosome;<br>• males affected/female carriers/occurs more in males; | | (2) |
| | (ii) | • blood fails to clot/excessive bleeding; | | (1) |

**Total for Question 9 = 10 marks**

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 6 of 7 · Question 5 part b.i · 2 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 5 (a) (i) | | • change in nucleotide/base/codon sequence;<br>• of DNA; | | 2 |
| | (ii) | • new/different (sequence of) amino acids;<br>• different protein / enzyme / different polypeptide<br>• different phenotype/enzyme does not bind to substrate; | | 3 |
| (b) (i) | | • allele for haemophilia on X chromosome/ not on Y chromosomes;<br>• hairy ears could only be males / haemophilia could occur in females (and males); | | 2 |
| | (ii) | • 1 = XX;<br>• 2 = $XY^H$; | | 2 |
| | (iii) | • chance of being a girl = 50%/ this can be indicated in a punnet square;<br>• girl will have XX genotype / girl indicated as XX from punnet square/ girls have two X;<br>• no Y chromosome so no chance of hairy ears; | | 3 |

**Total 12 marks**

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 6 of 7 · Question 5 part b.i · 2 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Explain sex-linked inheritance and why males are affected more

Answer 7 of 7 · Question 6 part a.iii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 (a) | (i) | 2500 males;<br>$\frac{2500 \times 0.8}{100}$;<br>= 20; | ecf if use 5000 as starting figure<br>40 scores 2 marks<br>any number×0.8=1 | 3 |
| | (ii) | small number/ not enough tested; | | 1 |
| | (iii) | • many more males than females with condition;<br>• show that it is sex-linked;<br>• to X chromosome;<br>• females are XX /males one X chromosome/XY<br>• females can be carriers( so fewer)/males cannot be carriers( so more);<br>• (relatively) small number (in whole population)/small chance of being colour blind;<br>• indicates that it is recessive; | | 5 |
| (b) | | parents $X^BY$ × $X^BX^b$;<br>gametes $X^B$ Y $X^B$ $X^b$;<br>offspring genotypes $X^BX^B$ $X^BX^b$ $X^BY$ $X^bY$;<br>offspring phenotypes normal carrier/ disease normal red-green;<br>female female male male | If parents are incorrect ecf for mp 2,3 and 4<br><br>If X and Y are not used- max of 1 mark | 4 |
| | | | | **Total = 13** |

## Type MS · Explain sex-linked inheritance and why males are affected more

### Answer 7 of 7 · Question 6 part a.iii · 5 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |